



Integrated Antiship Missile Project

Solid-propellant sizing, DATCOM aerodynamics, 6DOF dynamics, autopilot design and closed-loop guidance (MANSUP / Exocet MM40 class)

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1 Project Overview

This report consolidates the full antiship missile design into a single, closed-loop engagement. The vehicle is a solid-propellant, sea-skimming cruise missile equivalent to the Exocet/MM40 (MANSUP) class. Starting from a preliminary sizing, the model is carried through DATCOM aerodynamics, a variable-mass 6DOF model, gain-scheduled autopilots and, finally, a complete 6DOF guided flight (`GuiamentoControle.slx`).

Relative to the earlier integrated report, the guidance is no longer studied in a decoupled horizontal point-mass projection. It is now flown in the true vertical-plus-lateral geometry, with proportional navigation projected onto the body axes and gravity compensation. Because the projections changed, the numerical results changed as well: the headline figures below come from the current closed-loop 6DOF model.

1.1 Concept of Operations (CONOPS)

Table 1: Flight profile of the sea-skimming engagement.

Phase	What happens
Launch and boost	Booster accelerates to $\approx$ Mach 0.93 in 4 s; the missile descends to skim altitude. Fins locked during the first T_{Trav} .
Low-level cruise	Sustainer holds Mach 0.90; altitude control keeps $\approx$ 10–26 m, below the radar horizon.
Terminal (homing)	Seeker locks the target; proportional navigation commands the intercept.
Bingo	Proximity fuze detonates at < 20 m from the ship.

1.2 Engineering Chain

The missile is defined once and that definition feeds every stage consistently. The single source of truth is `00_Comum/DadosMissilComum.m` and `DadosCGInerciaComum.m`; each stage folder calls them through a forwarder.

Sizing $\rightarrow$ Aerodynamics (DATCOM) $\rightarrow$ 6DOF dynamics $\rightarrow$ Autopilots $\rightarrow$ Guidance & Control

The final stages are integrated in a single Simulink model, `GuiamentoControle.slx`, whose top level is shown in Figure 1. It closes the loop between six blocks: the *Target* (the ship), the *Seeker* (AD, which measures the line-of-sight geometry), *Guidance & Control* (cruise attitude control and terminal proportional navigation), the *Autopilot* (which drives the fins), the *6DOF Dynamics* (the variable-mass rigid-body airframe), and *Flight Termination* (which stops the run on bingo, sea impact, target overrun or low speed). Each block feeds the next, and the complete missile state is fed back every integration step.

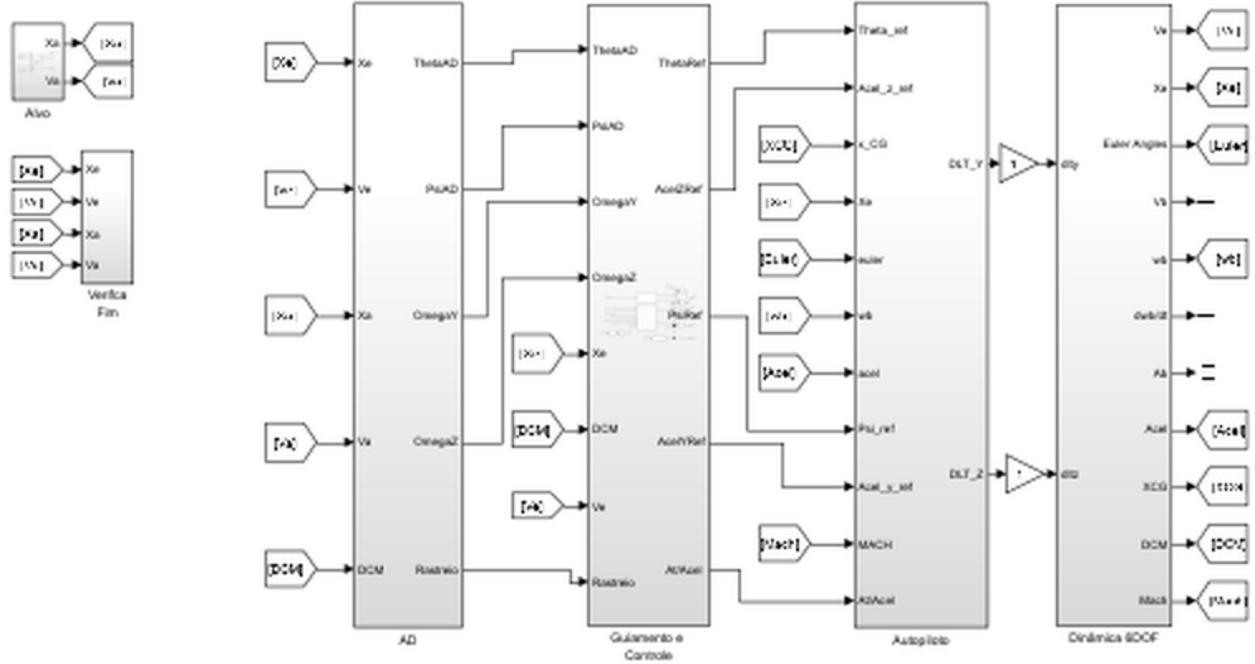


Figure 1: Top level of the integrated closed-loop Simulink model `GuiamentoControle.slx`: Target, Seeker (AD), Guidance & Control, Autopilot, 6DOF Dynamics and Flight Termination.



1.3 Design Requirements

Table 2: Preliminary design requirements.

Parameter	Symbol	Value	Role
Length	L	5.79 m	airframe geometry
Diameter	D_{ref}	0.35 m	reference area
Range	—	70 km	operational requirement
Cruise speed	—	Mach 0.90	high-subsonic limit (avoids shock wave)
Warhead	CDG	165 kg	payload / lethality
Skim altitude	H_{ref}	≈ 10 m	stealth (sea-skim)
Terminal maneuver	—	$\geq 5g$	intercept
Airframe bandwidth	—	≥ 0.5 Hz	control agility

Mach 0.90 is chosen because above it the flow goes transonic: a shock wave forms, drag spikes and the missile burns fuel and loses speed. Cruising at the top of the subsonic band is the fastest possible without paying the shock-wave penalty.

1.4 Headline Result

The closed-loop 6DOF engagement reaches the ship with a miss distance of 17.9 m (below the 20 m fuze radius) at Mach 0.90, $t \approx 207$ s. All requirements are satisfied and verified in simulation.



2 Propulsion — Two-Stage Solid

Propulsion is solid only, in two pulses (`dimensionamento_exocet.m`, parameters in `DadosMissilComum.m`).

- **Booster — star grain.** High burn-area/volume ratio gives high, short thrust; it takes the missile from rest to $\approx$ Mach 0.93. $M_{pB} = 125$ kg, $F_B \approx 67.4$ kN, $T_{qB} = 4$ s, $I_{sp} \approx 220$ s.
- **Sustainer — end-burner grain.** Burns only on the face (constant area) giving low, long thrust; it holds cruise for ≈ 229 s. $M_{pS} = 330$ kg, $F_S = 3.25$ kN, $T_{qS} = 228.6$ s, $I_{sp} \approx 240$ s.

2.1 Critical Decision — Sizing the Sustainer

The classic mistake is to size the sustainer for the drag at $\alpha = 0$ ($C_D \approx 0.42 \Rightarrow F_S \approx 2.3$ kN). But the missile cruises at a *trim* angle of attack, where the real drag rises to $C_{D,trim} \approx 0.58$. Sized for $\alpha = 0$, thrust $<$ drag, and the missile decelerates and dives. The sustainer is therefore sized for the cruise drag with lift (trim):

$$F_S = \bar{q} S C_{D,trim}, \quad \bar{q} = \frac{1}{2} \rho V^2, \quad V = 0.9 a, \quad M_{pS} = \frac{F_S T_{qS}}{I_{sp} g}.$$

With $C_{D,trim} \approx 0.58$ this gives $F_S = 3.25$ kN: thrust equals drag at Mach 0.90 and cruise holds.

Table 3: Thrust and propellant schedule `D.VProp` = [t F M_{prop}].

t (s)	Thrust (N)	M_{prop} (kg)
0.0	67 380	455
4.0	67 380	330
4.1	3 250	330
232.7	3 250	0
232.8	0	0

The launch mass is $M_0 \approx 930$ kg (empty 475 kg plus propellant 455 kg), and the maximum simulation time is set past the sustainer burnout.



3 Mass, CG, Inertia and Static Margin

The structural layout is defined nose to tail in `DadosCGInerciaComum.m`. The convention is $x = 0$ at the nose tip, positive forward, so all section positions are negative.

Table 4: Structural layout (nose to tail).

Section	Length	Mass	Role
Cont (guidance/nose)	1.09 m	201 kg	seeker + ballast to move CG forward
CDG (warhead)	0.90 m	165 kg	payload
Sustainer	2.35 m	60 + 330 prop	cruise motor
Booster	1.25 m	35 + 125 prop	acceleration motor
Nozzle	0.20 m	14 kg	—

The launch mass is $M_0 \approx 930$ kg (empty 475 kg + propellant 455 kg). The CG migrates forward as propellant burns and is recomputed in five burn phases: $x_{cg,0} = -2.65$ m $\rightarrow$ -1.70 m at burnout. Inertia is recomputed per phase (I_{xx} from a solid cylinder, $I_{yy} = I_{zz}$ by parallel axis over the sections).

3.1 Static Margin and Maximum Maneuver (Gmax)

The static margin is the distance in calibers between CG and neutral point:

$$\text{margin} = \frac{x_{CG} - x_{CP}}{D_{ref}}.$$

More margin means more stability but less maneuverability. With $x_{cg,0} = -2.65$ m the margin is ≈ 1.1 cal. A more rearward CG (-2.75 m, ≈ 0.79 cal) forces trim to $\alpha \approx -21^\circ$, outside $\pm 20^\circ$, so the fin cannot trim it; moving the CG forward to -2.65 m drops trim to $\approx -18^\circ$ and every burn phase closes within $\pm 20^\circ$.

For each CG phase and Mach, the maximum maneuver is found by solving $C_m = 0$ at maximum fin (20°) for the trim α , then converting to g :

$$G_{max} = \frac{\bar{q} S C_N(\alpha_{trim})}{m g}.$$

The result ranges $2 \rightarrow 7.4g$ across CG phases; the terminal value is $\approx 5.6g \geq 5g$.

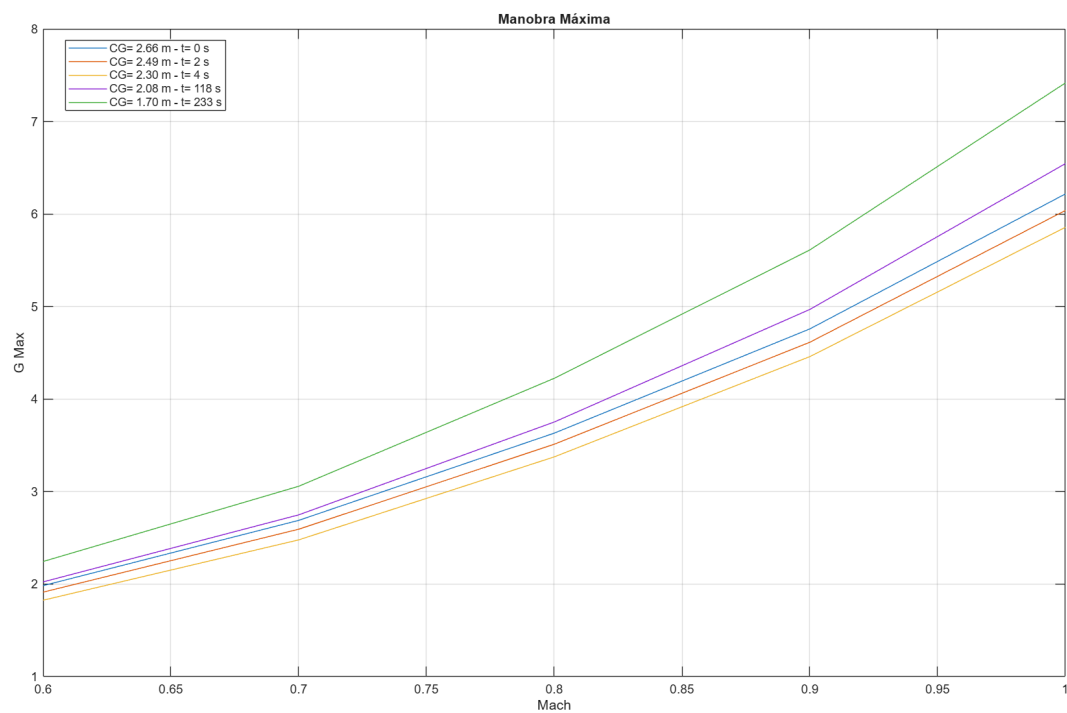


Figure 2: Maximum maneuver G_{max} versus Mach for the five CG phases. All curves trim within $\pm 20^\circ$.



4 Aerodynamics — DATCOM Database

The aerodynamic database is generated by Missile DATCOM (semi-empirical) through the course pipeline in `RunDatcom2/`:

`for005_mansup.dat` $\rightarrow$ `RunDatcom2.exe` $\rightarrow$ `Coef_mansup.dat` $\rightarrow$ `Coef2Mat.m` $\rightarrow$ `M_aed.mat`.

`M_aed.mat` holds three-dimensional tables in (fin deflection δ , Mach, α). The routine `calcula_coef( $\delta$ ,Mach, $\alpha$ )` interpolates these tables and computes the stability derivatives by finite differences, bridging the aerodynamic database to the transfer functions.

The physical shape of the coefficients is:

- C_N versus α : linear (lift);
- C_m versus α : negative slope, i.e. statically stable;
- C_D versus α : parabolic (drag bucket at $\alpha = 0$).

The DATCOM moment reference (CRM) is stored with the table. The 6DOF model transfers the aerodynamic moments from the CRM to the instantaneous real CG, so the CG may differ from the DATCOM reference without re-running DATCOM (Section 5).

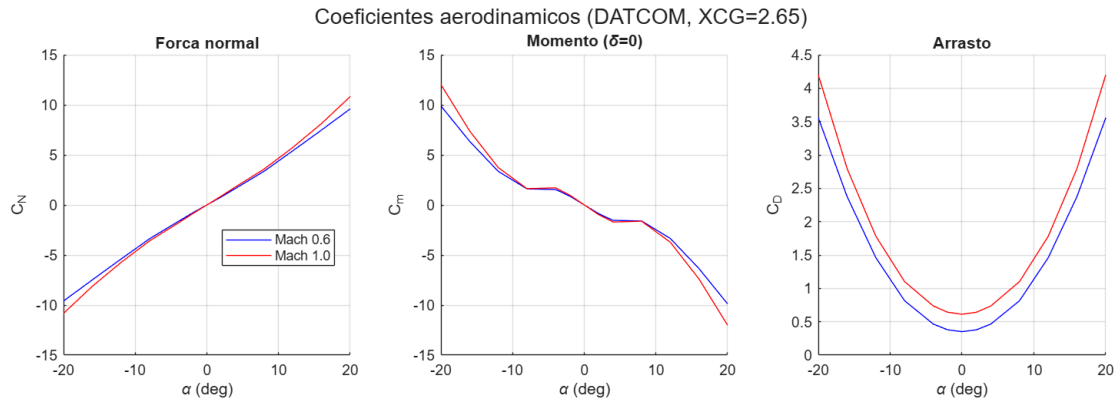


Figure 3: Aerodynamic coefficients C_N , C_m and C_D versus α .



5 6DOF Dynamics and Transfer Functions

The 6DOF model (`Dinamica6DOF.slx` and, in closed loop, `GuiamentoControle.slx`) integrates the six rigid-body equations with variable mass and inertia, summing aerodynamic, propulsive and gravitational forces and moments in the body frame.

5.1 Short-Period Linearization

`FTransDin.m` linearizes the pitch (short-period) motion about $\alpha = 0$, producing four second-order transfer functions with input equal to the fin deflection δ :

Table 5: Short-period transfer functions.

Transfer function	Output	Use
$\delta \rightarrow q$	pitch rate	inner rate-gyro loop
$\delta \rightarrow \theta$	attitude ($= q \cdot 1/s$)	outer attitude loop
$\delta \rightarrow \alpha$	angle of attack	—
$\delta \rightarrow A_z$	lateral acceleration	acceleration loop

The moment transfer from the DATCOM reference to the real CG is

$$\Delta x_{CG} = \frac{x_{CG} - x_{CRM}}{D_{ref}} \Rightarrow C_{m\alpha} = \frac{dC_M}{d\alpha} + \Delta x_{CG} C_{z\alpha}.$$

5.2 Why the Airframe Needs an Autopilot

The bare airframe is lightly damped: $\xi \approx 0.08$ in open loop, so a fin deflection excites an oscillation that takes long to decay, and $\omega_n \propto \sqrt{q}$ rises with speed. The acceleration channel additionally has a non-minimum-phase zero at $\approx +13.7$ rad/s, which produces the initial dip in the acceleration step and limits the achievable bandwidth. The inner rate-gyro loop feeds back q and raises the damping from $\xi \approx 0.08$ to ≈ 1 .

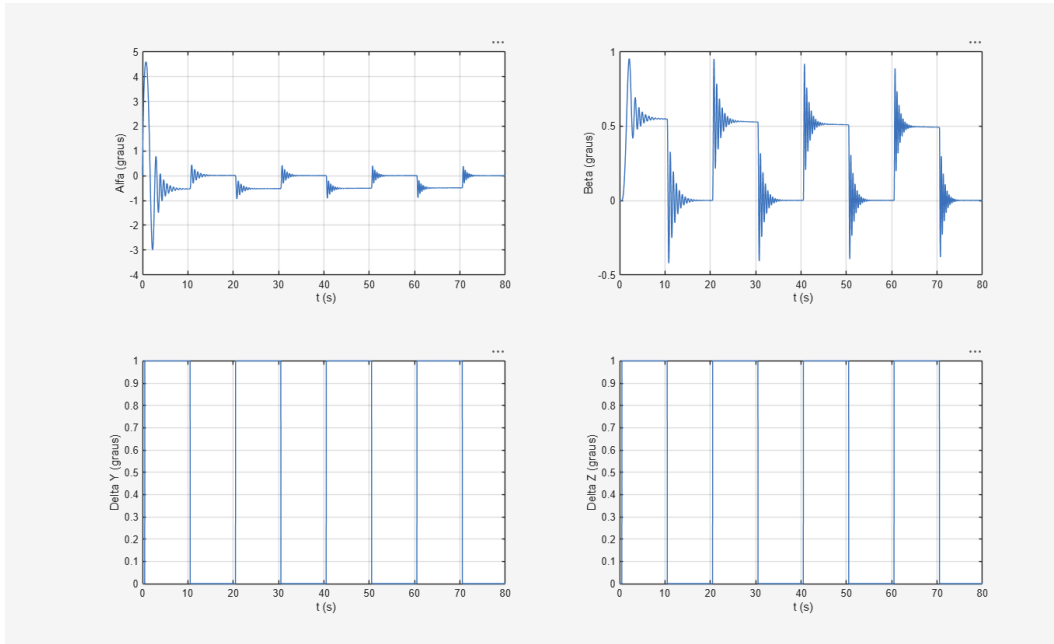


Figure 4: Open-loop free response of the airframe to a train of fin steps: α and β oscillate and decay slowly (short-period damping $\xi \approx 0.08$), with the frequency rising as the missile accelerates. This lightly damped behavior is what the autopilot must correct.



6 Autopilots — Two Loops, Root-Locus Design

Between the guidance command and the fins sits the autopilot. Two identical autopilots are used, one for pitch and one for yaw, each built as a nested pair of loops and designed by root locus (`ProjRLocus.m`). The three loops are:

1. **Inner (rate gyro):** feeds back the body rate q (or r) and damps the short-period. This is what turns the lightly damped airframe of Section 5 into a well-behaved plant, raising the damping from $\xi \approx 0.08$ to ≈ 1 .
2. **Outer attitude:** a proportional term plus an integrator tracking $q \rightarrow \theta$, with the gain chosen on the root locus where the dominant damping is $\zeta \approx 0.5$. It is used in cruise, driven by the altitude controller.
3. **Outer acceleration:** an integrator $-1/s$ tracking the lateral acceleration, with the highest gain that keeps the dominant damping $\zeta_{dom} \geq 0.6$. The gain is bounded by the non-minimum-phase zero of the acceleration transfer function. It is used in the terminal phase, driven by proportional navigation.

The gains are selected on the root locus of each loop, sweeping the design across the full flight condition (Section 6.2).

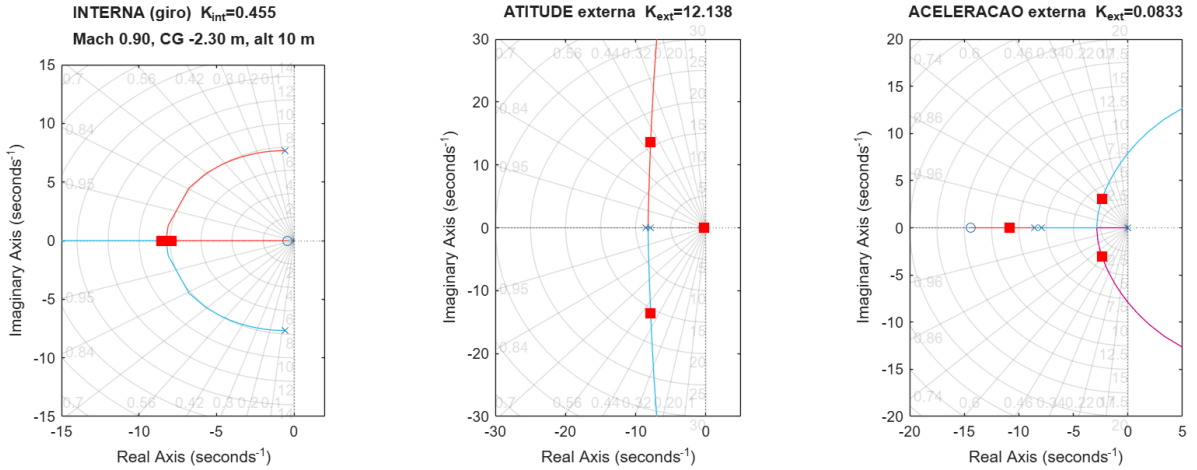


Figure 5: Root locus of the three loops; the square marks the selected gain.

6.1 Autopilot Model in Simulink

The autopilot is implemented as the `Autopiloto` subsystem of `GuiamentoControle.slx`. Figure 6 shows the test harness used to design and verify it in isolation: pulse generators inject attitude/acceleration commands, the autopilot drives the fin deflections δ_y and δ_z , and the 6DOF block returns the airframe response, which is fed back to the autopilot.

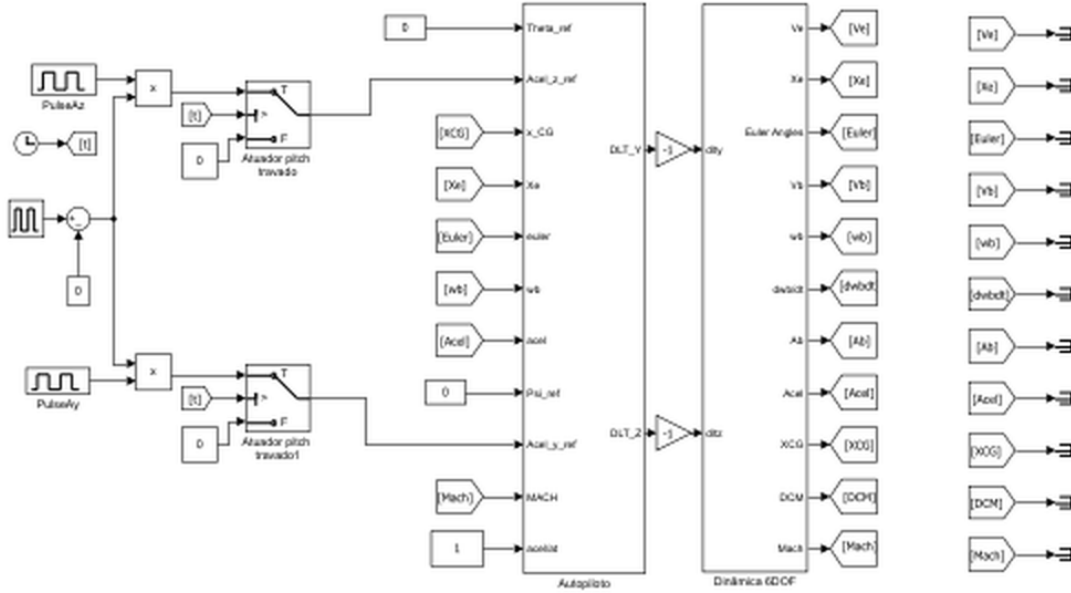


Figure 6: Simulink test harness for the autopilot: command generators (left) drive the Autopiloto subsystem, which commands the Dinâmica 6DOF block; the response is fed back. The *actuator-lock* switches (*Atuador pitch travado*) force $\delta = 0$ during the boost.

Internally, the Autopiloto subsystem is organized in clearly separated sections. Each channel (pitch/yaw) contains an attitude tracker and an acceleration tracker in parallel, and a switch selects which one is active according to the flight mode. Figure 7 shows the architecture of a single channel and Table 6 describes each section.

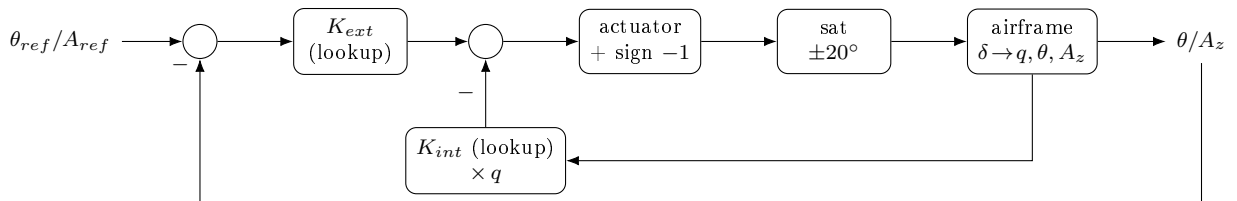


Figure 7: Architecture of one autopilot channel: outer loop (gain-scheduled K_{ext} , attitude or acceleration), inner rate-gyro loop (gain-scheduled K_{int} on q), actuator with sign inversion and $\pm 20^\circ$ saturation, and the airframe transfer functions.

Table 6: Sections of the **Autopiloto** subsystem and their function.

Section / block	Function
Demux of inputs	Splits the incoming vectors into the signals each loop needs: attitude θ, ψ from euler , rates q, r from wb , accelerations A_z, A_y from acel , and the scheduling variables (altitude, Mach, x_{CG}).
Actuator-lock switch	During the first $T_{T_{rav}}$ (boost) it forces $\delta = 0$: the missile is still accelerating and the fins are held. After $T_{T_{rav}}$ the command passes through.
Attitude tracker	Outer loop for cruise. Computes the attitude error $\theta_{ref} - \theta$, applies the gain-scheduled external gain, and subtracts the inner rate-gyro signal. Used with the sea-skim altitude controller.
Acceleration tracker	Outer loop for homing. Same inner structure, but with an integrator $1/s$ on the acceleration error so the steady-state error is zero. Used with proportional navigation.
Mode selector (switch)	Selects the active tracker (attitude vs. acceleration) from the At/Acel flag set by Guidance & Control.
Gain-scheduled lookups	Each loop reads its gain from a 3D table $T(x_{CG}, \text{alt}, \text{Mach})$ interpolated in real time (Section 6.2).
Actuator + sign inversion	The commanded deflection passes through the actuator with a -1 gain, correcting the sign convention of the $\delta \rightarrow q$ transfer function (Section 6.3).
Saturation $\pm 20^\circ$	Hard mechanical limit of the fin; the autopilot output can never exceed it, independently of the command.

6.2 Gain Scheduling

A single fixed set of gains cannot fly the whole mission, because the plant changes with the flight condition. The gains vary strongly with *Mach* (through the dynamic pressure $\bar{q}$, which scales all aerodynamic forces) and with the *CG position* (through the static margin, which changes as propellant burns), and only weakly with *altitude*. The design therefore sweeps six CG points (the five burn phases), two altitudes and three Mach numbers, and stores the resulting gains in three-dimensional tables $T_{xxx}(x_{CG}, \text{alt}, \text{Mach})$, saved to **autopiloto.mat**.

At run time, each autopilot loop reads its gain by interpolating (**interp**) its table at the instantaneous flight condition. The scheduling grid uses $\text{Mach} = [0.6 \ 0.75 \ 0.9]$ and $\text{alt} = [10 \ 100]$ m. During the initial low-speed transient (below Mach 0.6) the tables extrapolate, which is visible as a short transient in the flight time series but does not affect the cruise or terminal behavior.

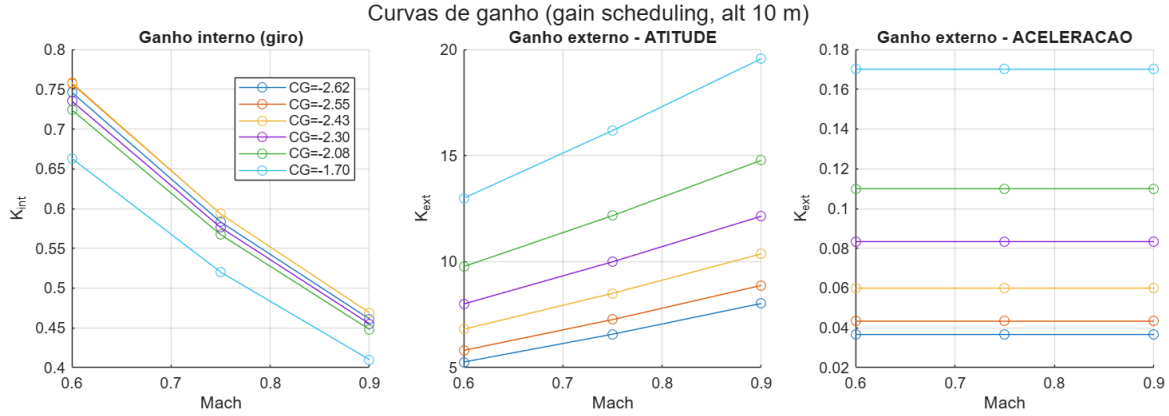


Figure 8: Gain schedule: internal and external gains versus Mach for each CG phase.

6.3 Sign-Convention Correction

This missile's $\delta \rightarrow q$ transfer function has *negative* gain, the opposite convention to the anti-drone reference model the project started from. If nothing is done, the nominally negative feedback becomes positive: the loop destabilizes and the executed motion comes out inverted with respect to the command. This is not a tuning issue but a sign issue, so it must be fixed structurally.

Two coordinated corrections restore negative feedback and keep the design and the simulation consistent. In the gain design, the actuator enters the loop with a negated numerator (`num_at = -AT.num`). In the simulation model, a Gain = -1 is placed on each autopilot deflection output (pitch and yaw); this same inversion also reconciles the DATCOM aerodynamic sign convention with the Simulink coordinate system. All four sign combinations (actuator $\pm$, gains $\pm$) were tested, and only the pair “actuator negated with positive gains” is stable, which is the one adopted.

6.4 Performance — Step Responses and Command Tracking

The closed-loop autopilot is verified against the bandwidth requirement with attitude, acceleration and $5g$ steps, and then confirmed inside the full 6DOF simulation.

- **Attitude step:** DC gain = 1.00, $\approx 8\%$ overshoot, bandwidth 2.6–3.6 Hz. The slow tail comes from the incidence zero.
- **Acceleration step:** DC gain = 1.00, an initial dip caused by the non-minimum-phase zero, $\approx 8\text{--}10\%$ overshoot, bandwidth 0.7–1.3 Hz.

A DC gain of 1.00 means the output settles exactly on the command — the autopilot follows what it is told. Both channels clear the 0.5 Hz requirement.

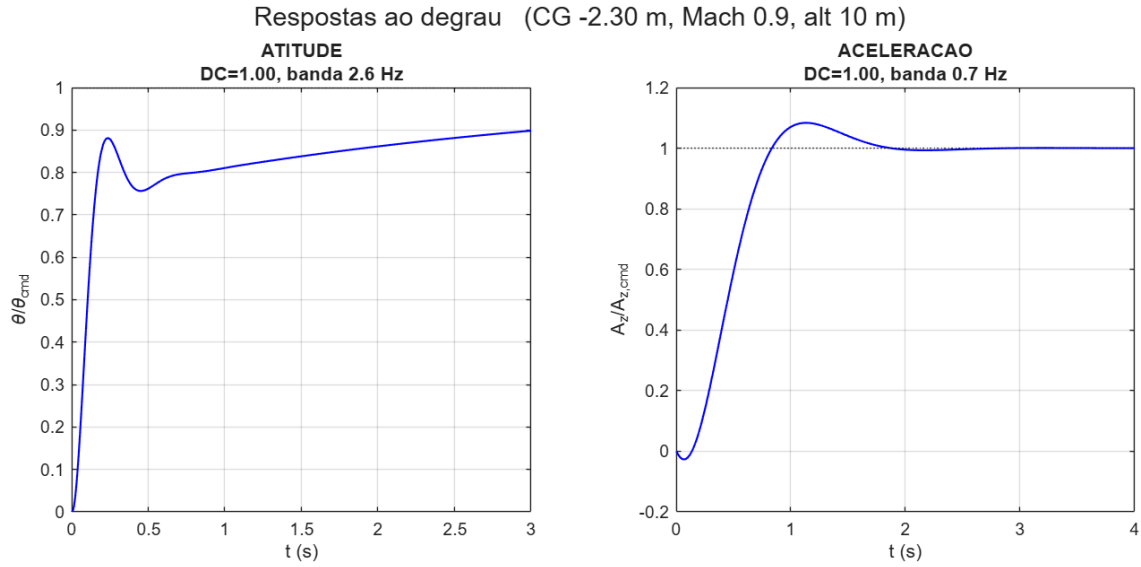


Figure 9: Attitude and acceleration step responses.

For the sizing $5g$ command, the simulated response reaches $5g$ faster than the 0.5 Hz bandwidth requirement (it overshoots to $\approx 5.48g$ and settles at $5g$), so the same design satisfies both the maneuver and the bandwidth requirements at once.

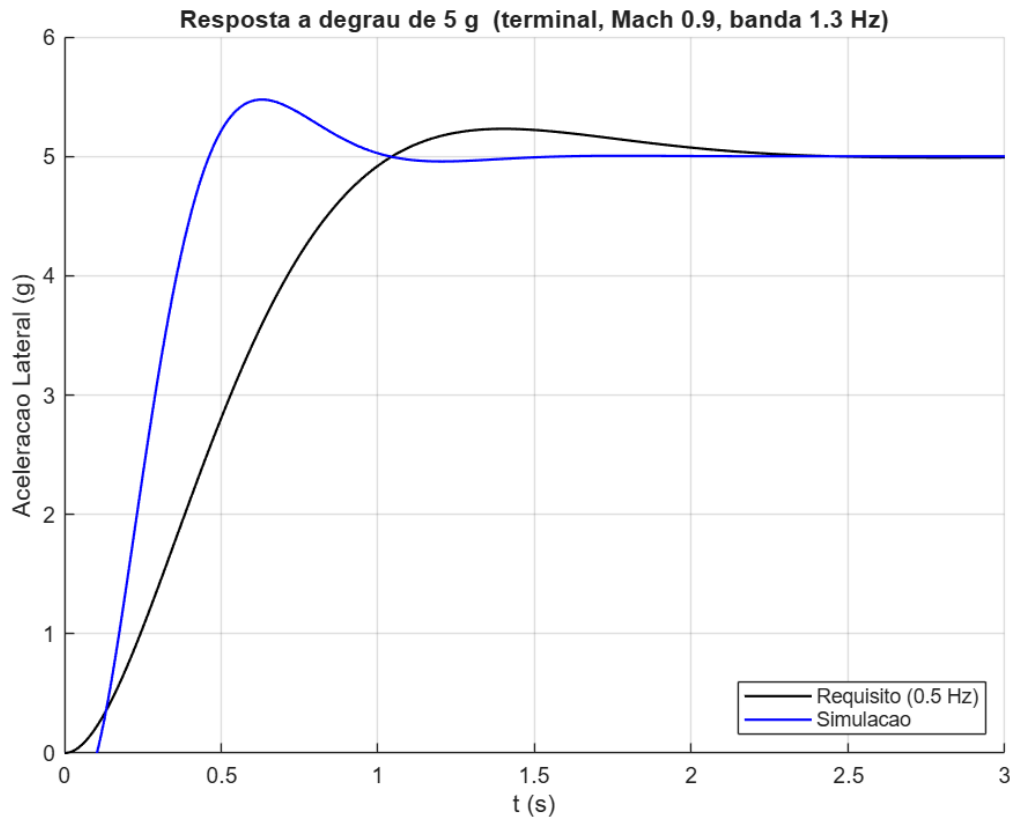


Figure 10: $5g$ step response against the bandwidth requirement.

Finally, in the complete 6DOF simulation the commanded and executed responses (θ , ψ , A_y , A_z) practically coincide (Figure 11), confirming that the scheduled autopilot tracks the guidance command across the whole flight, not only at the isolated design points.

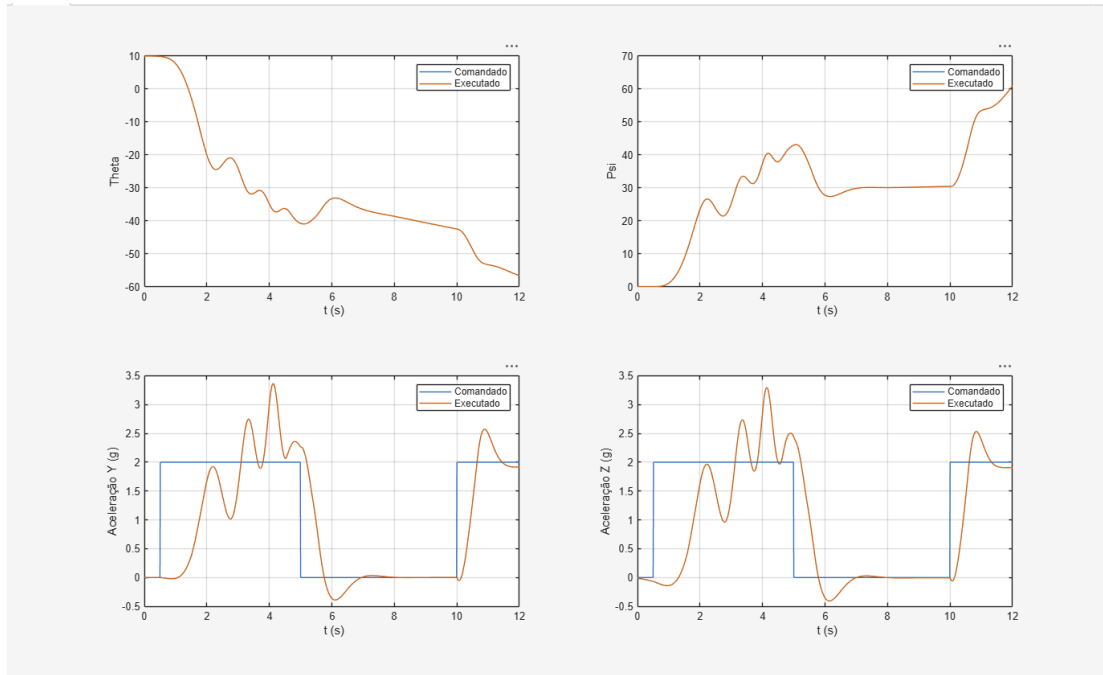


Figure 11: Full 6DOF: commanded versus executed responses (θ , ψ , A_y , A_z); the curves are nearly indistinguishable.



7 Control and Guidance

7.1 Altitude Control — Sea-Skimming

Low-level flight is a proportional loop from altitude error to attitude command; the resulting θ_{ref} is tracked by the attitude autopilot:

$$\theta_{ref} = \text{sat}_{\pm 20^\circ} [K_{alt} (H_{ref} - h)].$$

The altitude gain is $K_{alt} = 1^\circ/\text{m}$ (tuned; below this value the missile dives), with a reference $H_{ref} = 10$ m and a saturation on the climb angle. A proportional law is enough here because the inner attitude autopilot already provides the damping and the steady tracking: the altitude loop only has to bias the nose up or down, and the saturation prevents an absurd command (for instance, “dive 90° ” just because the missile is momentarily too high). The climb-angle limit also keeps the trajectory shallow, which is what makes the flight sea-skimming rather than a series of dives and climbs.

The result is a three-level cascade: altitude control feeds the attitude autopilot, which in turn relies on the rate-gyro loop. In flight the skim altitude settles at ≈ 26 m during cruise, safely below the ship’s radar horizon. During the first T_{Trav} the fins are locked ($\delta = 0$) and θ_{ref} is forced to zero, since the missile is still accelerating on the booster and has neither the airspeed nor the control authority to maneuver.

7.2 Terminal Guidance — Proportional Navigation

In the terminal phase the missile uses proportional navigation: the commanded acceleration is proportional to the line-of-sight (LOS) rotation rate, projected onto the body axes and gravity-compensated:

$$\vec{a}_{cmd} \approx \underbrace{N V \dot{\sigma}_{LOS}}_{\text{PN law}} \cdot \underbrace{\cos \theta_{AD}, \cos \psi_{AD}}_{\text{body-axis projection}} + \vec{g}_{comp}.$$

The chain is: the seeker (AD Ideal) measures LOS angles and rates (atan2 of the relative position; cross product over range for the rate); the terminal guidance forms $N V \dot{\sigma}_{LOS}$, projects it onto the body axes and adds the gravity-compensation term (gravity rotated into the body frame through the DCM); the autopilot in acceleration mode turns the A_z/A_y references into fin deflections. The gravity-compensation term is what keeps the missile from sagging during the terminal maneuver — a piece the flat point-mass projection did not require but the true-geometry 6DOF flight does.

Table 7: Guidance and seeker parameters.

Parameter	Value
Navigation constant N	3.5
Seeker turn-on time	$t = 10$ s
Seeker gimbal cone	$\pm 40^\circ$
LOS-rate limit	$\pm 40^\circ/\text{s}$
Tracking-range saturation	500 m
Proximity fuze radius	20 m
Minimum speed V_{min}	170 m/s

Before seeker turn-on the missile flies pre-programmed by heading (Rumo0) and altitude. The stop conditions are bingo, ground/sea impact, target overrun (miss), and speed below V_{min} .



8 Results

8.1 Requirements Check

Table 8: Requirement verification in the closed-loop 6DOF simulation.

Criterion	Required	Obtained	Status
Bingo (hits target)	miss < 20 m	17.9 m, Mach 0.90, 207 s	Satisfied
Terminal maneuver	$\geq 5g$	$\approx 5.6g$ ($G_{\max} 2 \rightarrow 7.4g$)	Satisfied
Airframe bandwidth	≥ 0.5 Hz	attitude 2.6–3.6 Hz, accel 0.7–1.3 Hz	Satisfied
Trim (all CG phases)	within $\pm 20^\circ$	launch -18° ... terminal -12°	Satisfied
Follow command	DC = 1, stable	DC = 1.00, damped	Satisfied
Cruise held	Mach 0.90	sustainer holds (thrust = trim drag)	Satisfied

8.2 Homing Flight

The horizontal trajectory reaches the target, the altitude profile is sea-skimming (≈ 26 m), and the speed holds cruise. The miss distance is 17.9 m < 20 m, so the fuze detonates (BINGO). With the target nearly ahead, the LOS rate is small, so the terminal maneuver is small and the run is almost straight, with the boresight error tending to zero at impact.

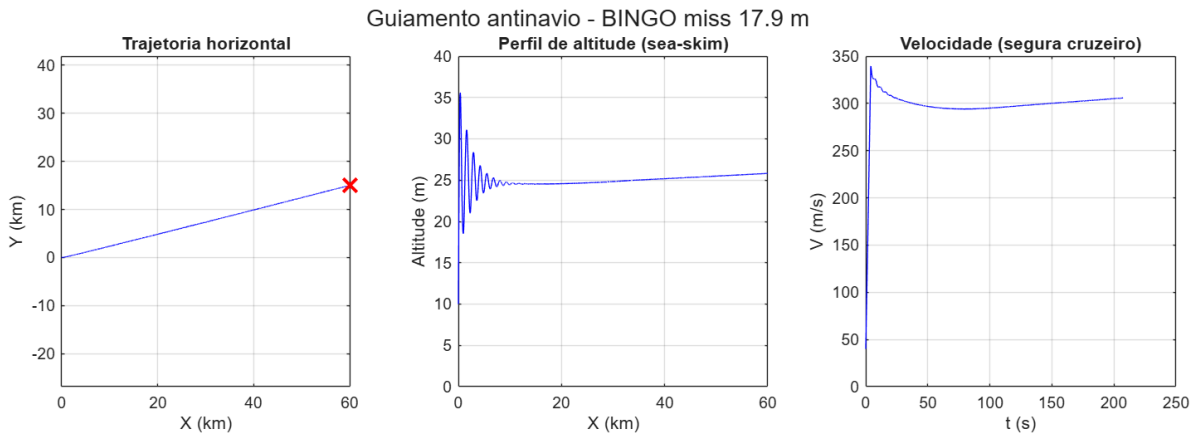


Figure 12: Homing trajectory to bingo (horizontal trajectory, altitude profile and speed).

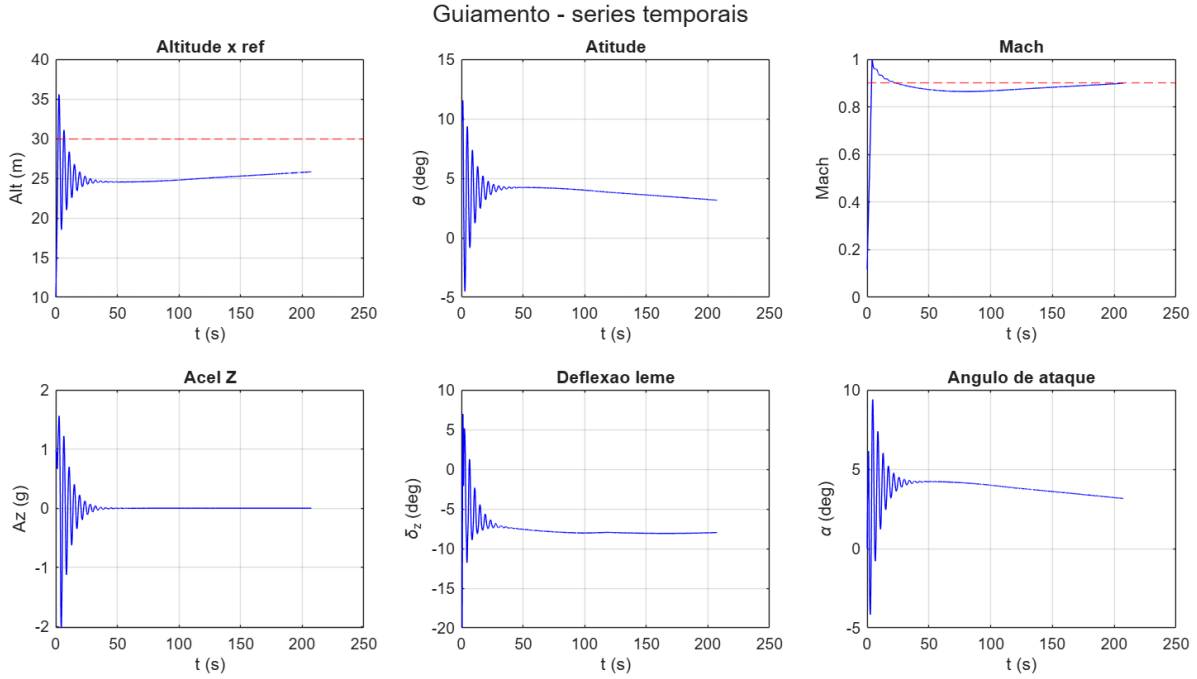


Figure 13: Flight time series: altitude versus reference, attitude θ , Mach, acceleration A_z , fin deflection δ and angle of attack α .

The flight phases are: boost 0–4 s ($0 \rightarrow \text{Mach } 0.93$); transient 0–40 s (gains extrapolate below Mach 0.6); smooth cruise 40–207 s; and terminal to bingo at 207 s.

8.3 Engagement Envelope

The envelope analysis (`Envelope6D0F.m`) sweeps launch positions and marks where the missile scores a bingo. Coverage is nearly omnidirectional (azimuths 0° – 150°), with a maximum range of ≈ 72 – 76 km > 70 km, limited by the sustainer burn (≈ 232 s). The missile fails only from directly behind (180°), as expected for an antiship weapon.

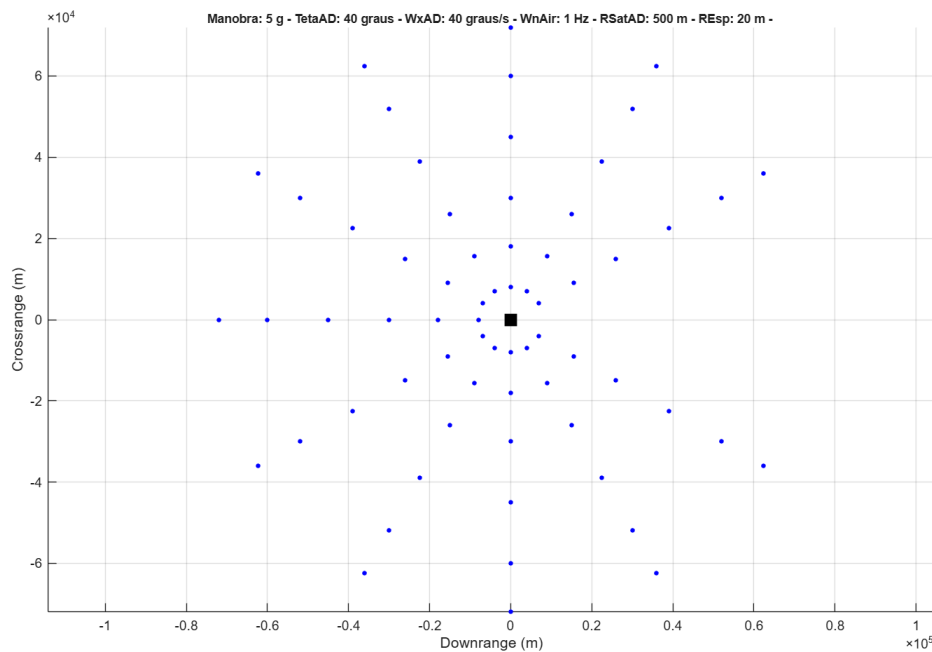


Figure 14: Engagement envelope (downrange versus crossrange, ship at the origin).



A bug was found during this validation: the pre-programmed cruise heading `C.Rumo0` was computed once (from the initial geometry) and stayed effectively at 0° . Beyond $\approx 40^\circ$ launch azimuth the missile crossed to the wrong side and the seeker could not recover, producing false “misses”. The fix is to recompute `Rumo0` from each launch geometry inside the loop, which corrected the whole envelope.



9 Modifications From the Reference Model

The project started from the coursework anti-drone / anti-UAV missile (laser-guided, air-to-air) and adapted it to the antiship role. The four main modifications are summarized below.

Table 9: Main modifications and their motivation.

Problem	Cause	Solution
Missile decelerated and dived Gmax curve did not close at launch	Sustainer sized for C_D at $\alpha = 0$ (under-sized) Rearward CG (-2.75 m), trim $\approx -21^\circ$, outside $\pm 20^\circ$	$F_S =$ cruise drag with trim ($C_D \approx 0.58$), holds Mach 0.90 CG forward (-2.65 m, 1.1 cal), trim -18° , all phases close
Command versus executed inverted	$\delta \rightarrow q$ with opposite sign to the reference model	Sign inversion in the design and in the simulation
A_z step over-oscillated; false envelope misses	Acceleration gain too aggressive; fixed cruise heading	Gain with $\zeta \geq 0.6$; heading re-computed per geometry



10 Integrated Interpretation and Course-Code Reuse

The stages are successive layers of the same preliminary missile.

- **Preliminary propulsion:** the sizing structure and `SimX` verification were retained; only the missile numbers changed.
- **Point-mass guidance:** the MANSUP Simulink model was the first guidance prototype (pre-programmed turn plus terminal switching) and remains the conceptual origin of the guidance logic now embedded in the 6DOF model.
- **DATCOM:** the `DATCOMreader/Coef2Mat` chain was retained; only the geometry input was rebuilt for the selected missile.
- **6DOF and autopilot:** the course dynamic model and root-locus method were reused with the new missile data, CG/inertia and aerodynamic matrix.
- **Guidance and control:** the anti-UAV `GuiamentoControle.slx` framework was adapted into a full antiship engagement, with sea-skim altitude hold, proportional navigation projected onto the body axes with gravity compensation, and an engagement envelope.

10.1 Single Source of Truth

All missile data live in `00_Comum/DadosMissilComum.m` and `DadosCGInerciaComum.m`; every stage folder calls them through a forwarder, guaranteeing that dynamics, autopilot and guidance refer to the same physical missile.

10.2 Integration Status and Limitations

The previous integration gap is now closed: the point-mass guidance and the 6DOF/autopilot models, previously in separate frameworks, are unified in a single closed-loop 6DOF mission (`GuiamentoControle.slx`). What remains preliminary is that propulsion, structural masses and aerodynamic geometry are engineering estimates, and the autopilot is verified at representative scheduling points together with a full homing flight and an engagement envelope, rather than over an exhaustively certified envelope. Within that scope the data flow is consistent: geometry, mass properties, aerodynamic coefficients and control gains all refer to the same missile configuration.



11 Reproducibility

Each stage is run in order, validating the output of one before moving to the next. The required tools are MATLAB with Simulink, the Control System Toolbox, the point-mass and 6DOF folders, and the DATCOM workflow (RunDatcom2, Coef2Mat.m).

Table 10: Execution sequence to reproduce the delivered results.

#	Stage	File / action	Output / check
1	Sizing	dimensionamento_exocet.m + SimX.mdl	mass, length, booster/sustainer, range and velocity plot
2	Point-mass guidance	run_mansup_cases.m	trajectories, switching times, miss distances
3	DATCOM geometry	write for005_mansup.dat	DATCOM input at the common moment reference
4	DATCOM processing	RunDatcom2 → Coef2Mat	Coef_mansup.dat, M_aed.mat
5	Coefficient plots	PlotCmCN.m, PlotCD.m, PlotCoef.m	aerodynamic coefficient figures
6	Max maneuver	PlotGMax.m	trim $\leq \pm 20^\circ$ and $\geq 5g$, all phases
7	6DOF open loop	VooUnico6DOF.m	short-period characterization
8	Autopilot design	ProjRLocus.m	autopiloto.mat, gains, root loci, steps
9	Altitude control	tune K_{alt} in ProjContAlt.m	stable sea-skim (≈ 26 m)
10	Full homing flight	RunGuiamentoFull.m (runs GuiamentoControle.slx)	trajectory, time series, miss 17.9 m
11	Engagement envelope	Envelope6DOF.m	hit zone, $\approx$ omnidirectional, 72– 76 km

11.1 Consistency Checks

Three checks keep the results comparable. First, the sizing values (diameter, length, mass, propulsion) must match the DATCOM geometry and the CG/inertia model. Second, DATCOM and 6DOF must share the same moment reference; the model transfers moments from the DATCOM reference to the real CG, so the CG may differ without re-running DATCOM:

```
Aer = load('M_aed.mat');
D.CRM = [-Aer.dados.XCG 0 0]';
```

Third, the missile is edited only in 00_Comum, which every stage folder forwards to.



12 Conclusions

All requirements were met and verified in a closed-loop 6DOF simulation:

- miss = 17.9 m < 20 m fuze $\rightarrow$ BINGO;
- Mach 0.90 cruise held (sustainer sized for the trim drag);
- terminal maneuver $\approx 5.6g \geq 5g$;
- bandwidth 0.7–3.6 Hz ≥ 0.5 Hz;
- trim within $\pm 20^\circ$ in every CG phase;
- engagement envelope up to 72–76 km > 70 km, nearly omnidirectional.

The main lessons learned are: size the sustainer for the trim drag, not for $\alpha = 0$; the $\delta \rightarrow q$ sign needs explicit correction; gain scheduling is essential across the full envelope; the Rumo0 bug affected the entire envelope until fixed; and the 00_Comum single source of truth is what kept the whole chain self-consistent. The complete chain, from sizing to guidance, is reproducible and self-consistent.